

Assignment 2: Design of Experiments and Evaluation of a Given System

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Introduction

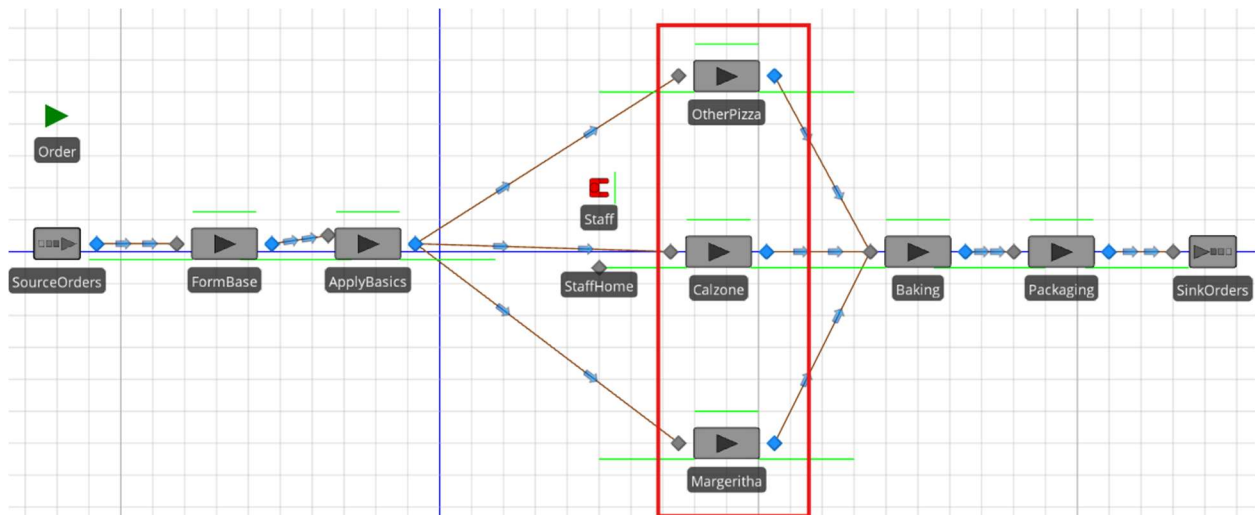


Figure 1: Model

The subject of our experiments is the Model in Figure 1, which consists of the following processes based on a fictional Pizza-Factory and consists of following stations: **Form_Base**, **Apply_Basic** (Toppings), special toppings (**Margherita**, **Other**, **Calzone**), **Baking** and **Packaging**. We require **Staff** to man the stations of special toppings (marked in red). The rest of the stations are automated. A table with the number of incoming orders at specific time during the day is given to simulate the demand.

Goal and Limitations

Our goal is to find the optimum combination, in other words minimum possible capacity of each of the automated station and staff in this model, to produce at least 500 Pizzas per day. The purpose is to help the owner of the Pizza Factory to identify and eliminate any other bottle necks in of the automated stations in this production line. Knowing that some orders will come in just before midnight and its production will be only done the next day, but this is acknowledged by the owner and totally fine, since it is a 24/7 production line.

Experiment plan

Scenario		Controls						Responses						
☐	Name	Status	FormBase_Capacity	ApplyBasics_Capacity	Staff_Capacity	Baking_Capacity	Packaging_Capacity	RateTable	Orders_Finished	Orders_NumberInSystem_Mean	Orders_NumberInSystem_Max	FormBaseMaxWaiting	ApplyBasicsMaxWaiting	SpecialToppingMaxWaiting
☒	Base_Scenario	Idle	1	1	2	3	1	OrderR...						
☒	Maximum_Scenario	Idle	100	100	100	100	100	OrderR...						
☒	Initial_Scenario	Idle	6	3	8	14	4	OrderR...						
☒	Scenario4	Completed	5	3	8	14	4	OrderR...	492.7	4.27033	32.2	5.5	2.1	
☒	Scenario5	Idle	6	2	8	14	4	OrderR...						
☐	Scenario6	Idle	6	3	8	14	4	OrderR...						
☐	Scenario7	Idle	6	3	8	14	4	OrderR...						
☐	Scenario8	Idle	6	3	8	14	4	OrderR...						
☐	Scenario9	Idle	6	3	8	14	4	OrderR...						
☐	Scenario10	Idle	6	3	8	14	4	OrderR...						
☐	Scenario11	Idle	6	3	8	14	4	OrderR...						
☐	Scenario12	Idle	6	3	8	14	4	OrderR...						

Figure 2: Experiment Matrix

The experiment Matrix (Figure 2) allows us to conduct experiments in multiple scenarios and is split in 3 parts:

1. Scenario, where we can index our experiment and see the progress of the simulation
2. Controls, where we can define our variables for our experiments
3. Responses, here we can define our KPIs and find the final KPI results of our experiments

Variables

Controls					
FormBase_Capacity	ApplyBasics_Capacity	Staff_Capacity	Baking_Capacity	Packaging_Capacity	RateTable

Figure 3: Variables

Since we only need to experiment with the automated stations and staff capacity we define the following variables: **FormBase_Capacity**, **ApplyBasics_Capacity**, **Staff_capacity**, **Baking_Capacity**, **Packaging_Capacity**. The 3 Specialty stations are not parameterized since they have infinite capacity, but are dependent on the number of staffs. We can ignore the **OrderRate** since it is always fix and the same rate of orders is used in all experiments.

KPIs

Responses							
Orders_Finished	Orders_NumberInSystem_Mean	Orders_NumberInSystem_Max	FormBaseMaxWaiting	ApplyBasicsMaxWaiting	SpecialToppingMaxWaiting	BakingMaxWaiting	PackagingMaxWaiting

Figure 4: KPI

Since we want to find the optimum throughput of the system within a day, we measure how many Orders are finished (**Orders_Finished**), additionally we want to measure the number of simultaneous orders within the system (**Orders_NumberInSystem_Mean**, **Orders_InSystem_Mean**) to get another more deterministic indication of the throughput of the system. For each of the measures above, we take the **max** occurrence to see the

behavior of the model during peak-demand time, and also the **average** for general statement about the system.

Additionally, we measure the max number of orders in the que of each station (**FormBaseMaxWaiting, ApplyBasicsMaxWaiting, SpecialToppingMaxWaiting, BakingMaxWaiting, PackagingMaxWaiting**) per iteration, to get an indication where the jam is and adjust the parameters. Thereby SpecialToppingMaxWaiting is the averaged sum of all 3 stations.

Implementation

Current situation and target situation

Scenario			Replic...	Controls					
☐	Name	Status	...	C...	FormBase_Capacity	ApplyBasics_Capac...	Staff_Capacity	Baking_Capacity	Packaging_Capacity
☐	Base_Scenario	Idle	10	0 of	1	1	2	3	1
☐	Maximum_Scenario	Idle	10	0 of	100	100	100	100	100

Figure 5: Baseline and Maximum Scenario

For the experiments we setup 1. the **Base_Scenario**, to confirm and validate the parameters with the real world and to get a base line for the following experiments. 2nd we setup another scenario to get the theoretical maximum for the KPIs, without any bottle necks in the system, we call it the **Maximum_Scenario**.

Scenario			Responses							
☐	Name	Status	Orders_Finished	Orders_NumberInSystem_Mean	Orders_NumberInSystem_Max	FormBaseMaxWaiting	ApplyBasicsMaxWaiting	SpecialToppingMaxWaiting	BakingMaxWaiting	PackagingMaxWaiting
☒	Base_Scenario	Completed	311.3	46.6828	238.7	231.4	1	1.5	3.9	2
☒	Maximum_Scenario	Completed	505	4.32719	31.1	1	1	1.8	1	1

Figure 6: Results of baseline and theoretical maximum

Initial Scenario

Now that we have the KPI results from the base line and theoretical optimum, we know what to aim for. To start somewhere, the time for process * 2, the thinking behind is, the longer it takes to process, the more capacity we need, and times 2 as a random factor.

FormBase: $3 \times 2 = 6$, **ApplyBasics:** $1.2 \times 2 = 3$, **SpecialTopping**(for **Staff_Capacity**): $((6+1)+5/3) \times 2 = 8$, **Baking:** $7 \times 2 = 14$, **Packaging:** $2 \times 2 = 4$

Scenario		Controls				
☐	Name	FormBase_Capacity	ApplyBasics_Capacity	Staff_Capacity	Baking_Capacity	Packaging_Capacity
☐	Base_Scenario	1	1	2	3	1
☐	Maximum_Scenario	100	100	100	100	100
☐	Initial_Scenario	6	3	8	14	4

☐	Name	Status	RateTable	Orders_Finished	Orders_TimeInSystem_Mean (Hours)	Orders_TimeInSystem_Maximum (Hours)	Orders_NumberInSystem_Mean	Orders_NumberInSystem_Max	Staff_Utility_Average	FormBaseMaxWaiting	ApplyBasicsMaxWaiting
☐	Base_Scenario	Completed	OrderRate	311.3	1.64127	4.52253	46.6828	238.7	0.546582	231.4	1
☐	Maximum_Scenario	Completed	OrderRate	505	0.205554	0.284937	4.32719	31.1	0.878352	1	1
☐	Initial_Scenario	Completed	OrderRate	504.4	0.206376	0.29971	4.34146	32.3	0.848032	3.5	2.7

Figure 7: Initial Scenario and results

As expected, after we factor in the time to process, the system compensate slow processing time with additional capacity. And the result shown in (Figure 7: Initial Scenario and results) of the initial scenario compared to the maximum scenario looks already promising.

Station by Station Optimization

To see whether this is the real optimum or whether there is another cheaper combination which can achieve the result (>500 Orders per day). We add a series of scenarios with lower capacities.

Scenario			Controls					
☐	Name	Status	FormBase_Capacity	ApplyBasics_Capacity	Staff_Capacity	Baking_Capacity	Packaging_Capacity	RateT...
☐	Base_Scenario	Idle	1	1	2	3	1	Order...
☐	Maximum_Scenario	Idle	100	100	100	100	100	Order...
☐	Initial_Scenario	Idle	6	3	8	14	4	Order...
☐	Scenario4	Completed	6	3	8	14	4	Order...
☐	Scenario5	Completed	6	2	8	14	4	Order...
☐	Scenario6	Completed	6	2	8	14	4	Order...
☐	Scenario7	Completed	6	2	8	14	4	Order...
☒	Scenario8	Completed	6	2	8	14	4	Order...
☒	Scenario9	Completed	6	2	8	14	4	Order...

Scenario			Responses							
☐	Name	Status	Orders_Finished	Orders_NumberInSystem_Mean	Orders_NumberInSystem_Max	FormBaseMaxWaiting	ApplyBasicsMaxWaiting	SpecialToppingMaxWaiting	BakingMaxWaiting	PackagingMaxWaiting
☐	Base_Scenario	Idle	311.3	46.6828	238.7	231.4	1	1.5	3.9	2
☐	Maximum_Scenario	Idle	505	4.32719	31.1	1	1	1.8	1	1
☐	Initial_Scenario	Idle	504.4	4.34146	32.3	3.5	2.7	2.1	3.5	3.2
☐	Scenario4	Completed	492.7	4.27033	32.2	5.5	2.1	1.7	2.8	3.5
☐	Scenario5	Completed	500.1	4.509	33.5	4.5	8.7	1.66667	1.9	3.4
☐	Scenario6	Completed	499.2	4.46613	32.3	4.3	7.6	1.8	1.8	3.6
☐	Scenario7	Completed	489.5	4.34605	32.9	3.9	7.8	1.63333	2.5	3.1
☒	Scenario8	Completed	503	4.51569	31.7	4.1	7.5	1.6	2.1	4.6
☒	Scenario9	Completed	496.5	6.08316	49.1	4.2	8.7	1.73333	1.8	27.5

Figure 8: Station By Station reduction and results

In Figure 8: Station By Station reduction we reduce the capacity of each station by 1 and reset it, if the **Order_Finished** is below 500, and leave it if it's at least 500. As you can see, we did this for each station separately and come to the conclusion, that **Apply_Basics_Capacity (Scenario 5)** and **Packaging_Capacity (Scenario8)** is overexaggerated, since the **Order_Finished** is not below 500. For packaging we also confirm it with further reduction (to only a capacity of 2), that 3 is indeed the optimum.

Scenario		Controls				
☐ Name		FormBase_Capacity	ApplyBasics_Capacity	Staff_Capacity	Baking_Capacity	Packaging_Capacity
☐ Base_Scenario		1	1	2	3	1
☐ Maximum_Scenario		100	100	100	100	100
☐ Initial_Scenario		6	3	8	14	4
☐ Scenario4		5	3	8	14	4
☐ Scenario5		6	2	8	14	4
☐ Scenario6		6	2	8	14	4
☐ Scenario7		6	2	8	13	4
☐ Scenario8		6	2	8	14	3
☐ Scenario9		6	2	8	14	2
☐ Scenario10		6	3	7	13	3
☐ Scenario11		6	2	7	13	3

Scenario		Responses							
☐ Name		Orders_Finished	Orders_NumberInSystem_Mean	Orders_NumberInSystem_Max	FormBaseMaxWaiting	ApplyBasicsMaxWaiting	SpecialToppingMaxWaiting	BakingMaxWaiting	PackagingMaxWaiting
☐ Base_Scenario		311.3	46.6828	238.7	231.4	1	1.5	3.9	2
☐ Maximum_Scenario		505	4.32719	31.1	1	1	1.8	1	1
☐ Initial_Scenario		504.4	4.34146	32.3	3.5	2.7	2.1	3.5	3.2
☐ Scenario4		492.7	4.27033	32.2	5.5	2.1	1.7	2.8	3.5
☐ Scenario5		500.1	4.509	33.5	4.5	8.7	1.66667	1.9	3.4
☐ Scenario6		499.7	4.46613	32.3	4.3	7.6	1.8	1.8	3.6
☐ Scenario7		489.5	4.34605	32.9	3.9	7.8	1.63333	2.5	3.1
☐ Scenario8		503	4.51569	31.7	4.1	7.5	1.6	2.1	4.6
☐ Scenario9		496.5	6.08316	49.1	4.2	8.7	1.73333	1.8	27.5
☐ Scenario10		505.8	4.4171	31	3.7	2.9	2.3	3.9	5.1
☐ Scenario11		497.3	4.4484	31.2	3.7	7.1	1.73333	2.2	4.1

Figure 9: Circle back and Final Solution results

In the last 2 experiments we circle back to check, what if the reduction in Staff and Baking Capacity is actually caused by congestion in **ApplyBasics** (Scenario5) and we can reduce the capacity of all following stations (Scenario10, Staff and Baking), without impacting the amount of **Orders_Finished**, if we only increase back **ApplyBasics** to 3? And indeed we have even a higher number 505 finished Orders. To confirm, we reduce back the **ApplyBasics** to 2 to see whether our theory is correct (Scenario 11). In the Figure 9: Circle back and Final Solution results we can see, we can see that the reduction has in deed brought the number of finished Orders down to below 500. So the final constellation is **FormBase_Capacity:6, ApplyBasics_Capacity:3, Staff_Capacity:7, Baking_Capacity:13, Packaging_Capacity:3** in Scenario 10.

Evaluation of Results

Scenario		Controls				
☒ Name		FormBase_Capacity	ApplyBasics_Capacity	Staff_Capacity	Baking_Capacity	Packaging_Capacity
☐ Base_Scenario		1	1	2	3	1
☐ Maximum_Scenario		100	100	100	100	100
☐ Initial_Scenario		6	3	8	14	4
☐ Scenario4		5	3	8	14	4
☐ Scenario5		6	2	8	14	4
☐ Scenario6		6	2	7	14	4
☐ Scenario7		6	2	8	13	4
☐ Scenario8		6	2	8	14	3
☐ Scenario9		6	2	8	14	2
☐ Scenario10		6	3	7	13	3
☒ Scenario11		6	2	7	13	3

Scenario		Responses							
☒ Name		Orders_Finished	Orders_NumberInSystem_Mean	Orders_NumberInSystem_Max	FormBaseMaxWaiting	ApplyBasicsMaxWaiting	SpecialToppingMaxWaiting	BakingMaxWaiting	PackagingMaxWaiting
☐ Base_Scenario		311.3	46.6828	238.7	231.4	1	1.5	3.9	2
☐ Maximum_Scenario		505	4.32719	31.1	1	1	1.8	1	1
☐ Initial_Scenario		504.4	4.34146	32.3	3.5	2.7	2.1	3.5	3.2
☐ Scenario4		492.7	4.27033	32.2	5.5	2.1	1.7	2.8	3.5
☐ Scenario5		500.1	4.509	33.5	4.5	8.7	1.66667	1.9	3.4
☐ Scenario6		499.7	4.46613	32.3	4.3	7.6	1.8	1.8	3.6
☐ Scenario7		489.5	4.34605	32.9	3.9	7.8	1.63333	2.5	3.1
☐ Scenario8		503	4.51569	31.7	4.1	7.5	1.6	2.1	4.6
☐ Scenario9		496.5	6.08316	49.1	4.2	8.7	1.73333	1.8	27.5
☐ Scenario10		505.8	4.4171	31	3.7	2.9	2.3	3.9	5.1
☒ Scenario11		497.3	4.4484	31.2	3.7	7.1	1.73333	2.2	4.1

Figure 10: Comparison

In Figure 10: Comparison compare the results, before and after the optimization. We reduced the capacity in terms of Staff, Backing and Packing Station without reducing the **Orders_Finished** below 500. Although both scenarios (**Initial_Scenario** and **Scenario 10**) are similar, the MaxWaiting orders of each station in the model of **Scenario 10** varies from 2.3 minutes to 5.1 minutes and in the **Initial_Scenario** only from 2.1 minutes to 3.5 minutes, which indicates a better load distribution and therefore a more stable production. This confirms that the initial Scenario is in deed the optimum as far as capacity given in whole number allows.

Discussion and Conclusion

Since the target is at least 500 Pizzas per day, and the owner is not concerned with perfect equilibrium but cost, we suggest the following for improvement.

Increase the capacity of the stations in the system as follow:

FormBase:6, ApplyBasics:3, Staff:7, Baking:13, Packaging:3 (_Capacity respectively).

This can save more than the variant before the circle back:

FormBase:6, ApplyBasics:2, Staff:8, Baking:14, Packaging:3 (_Capacity respectively).